

```

function estimate = markovpi(Ntrials,d)
Nhits=0;
cords=[1,1];

for i = 1:Ntrials
    dx=d*(2*rand()-1); %dx = ran(-delta,delta)
    dy=d*(2*rand()-1); %dy = ran(-delta,delta)
    %checking if lands in helipad square
    if abs(cords(1)+dx)<1
        if abs(cords(2)+dy)<1
            cords(1) = cords(1) + dx;
            cords(2) = cords(2) + dy;
        end
    end
    %checking if lands in helipad circle
    if cords(1)^2 + cords(2)^2 < 1
        Nhits=Nhits+1;
    end
end
estimate = Nhits/Ntrials; %estimate of pi/4
end

```

```
markovpi(1000,0.2)
```

```
ans =
0.8580
```

```
markovpi(10000,0.2)
```

```
ans =
0.8006
```

```
markovpi(100000,0.2)
```

```
ans =
0.7803
```

```
markovpi(1000000,0.2)
```

```
ans =
0.7830
```

```

deltas=linspace(0.01,10,1000);
N=100; %100 estimates for each delta

```

```

estimates=zeros(1,N);
sd=zeros(1,length(deltas));

```

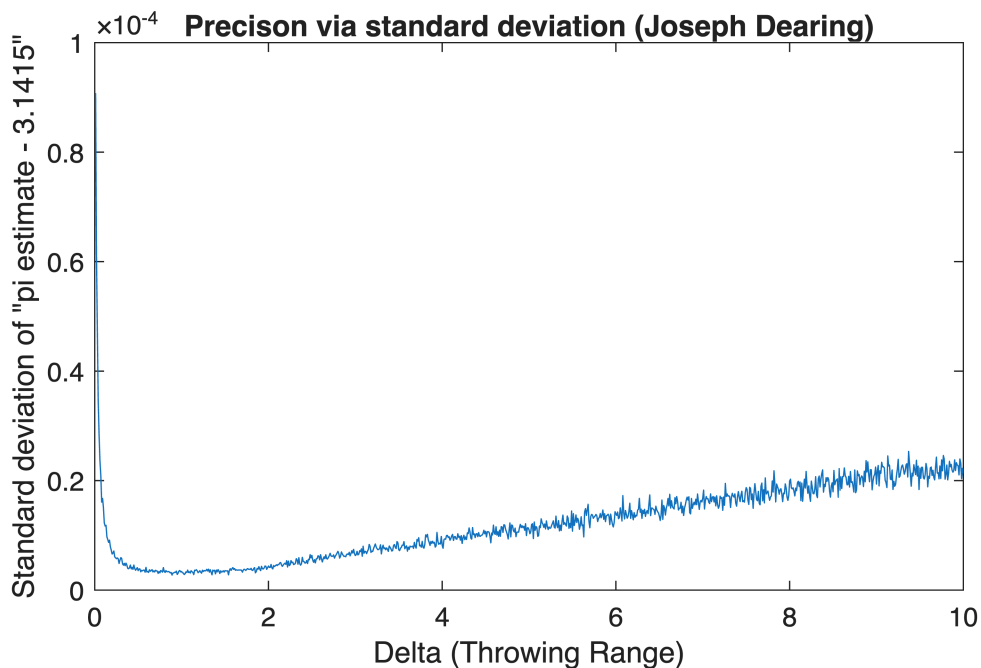
```

for i = 1:length(deltas)
    for j = 1:N
        estimates(j)=4*markovpi(10000,deltas(i))/10000;
    end
    sd(i) = std(estimates);
    %sd(x-c)=sd(x) for c constant hence no need to actually include 3.1415
end

```

%From graph we can see for delta very small we get high standard deviation,
 %likely due to the fact we are not traversing the entire helipad with the
 %amount of trials we have proposed. For delta approximately between 0.5 and
 %1.5 we get the lowest standard deviation hence the greatest precision in
 %estimating pi. As delta increases beyond these values the standard
 %deviation increases, due to the fact the higher we increase delta the
 %more
 %likely the stones will land outside the helipad.

```
plot(deltas,sd)
xlabel('Delta (Throwing Range)')
ylabel('Standard deviation of "pi estimate - 3.1415"')
title('Precision via standard deviation (Joseph Dearing)')
```



```
function estimate = metropolishastings(Ntrials,d)
Nhits = 0;
cords = [1,1];
h = sqrt(3)/2*d; %height of triangle

for i = 1:Ntrials

    %producing valid throw in triangle
    %i.e. sampling x'
    intri = false;
    while ~intri
        dx=d*(rand()-0.5);
        dy=d*(rand()-0.5);
        xtri = [cords(1),cords(1)+d/2,cords(1)-d/2];
        ytri = [cords(2)+h/2,cords(2)-h/2,cords(2)-h/2];
        if inpolygon(cords(1) + dx, cords(2) + dy, xtri, ytri)
            xprime=cords(1)+dx;
            yprime=cords(2)+dy;
```

```

        intri = true;
    end
end

%checking if in square and if reverse is possible
if abs(xprime)<1
    if abs(yprime)<1
        xtriprime = [xprime,xprime+d/2,xprime-d/2];
        ytriprime = [yprime+h/2,yprime-h/2,yprime-h/2];
        if inpolygon(cords(1), cords(2), xtriprime, ytriprime)
            %if true  $A(x',x)=1/(\text{area of equalateral triangle with side lengths delta})$ 
            %Since  $A(x,x')=1/(\text{area of equalateral triangle with side lengths delta})$ 
            %and  $\pi(x) = \pi(x')$  as we have uniform distribution, the
            %Metropolis-Hastings filter is one, hence the move is
            %accepted
            %if false  $A(x',x)=0$ 
            %therefore the Metroplois-Hastings filter is zero, hence
            %move is rejected.
            cords(1)=xprime;
            cords(2)=yprime;
        end
    end
end

if cords(1)^2 + cords(2)^2 < 1
    Nhits=Nhits+1;
end
end

estimate = Nhits/Ntrials; %estimate of  $\pi/4$ 
end

metropolishastings(1000,0.2)

```

```

ans =
0.4100

```

```
metropolishastings(10000,0.2)
```

```

ans =
0.7214

```

```
metropolishastings(100000,0.2)
```

```

ans =
0.7939

```

```
metropolishastings(1000000,0.2)
```

```

ans =
0.7992

```

```
function metropolis(d,t)
```

```

x=0;
samples=zeros(1,t);

pi24=@(x)exp(-x^2/2-x^4/2);

```

```

for i = 1:t

    dx=d*(2*rand()-1);
    r=pi24(x+dx)/pi24(x);
    upsiion=rand();

    if upsiion<min(1,r)
        x=x+dx;
    end

    samples(i)=x;

```

```

end

```

```

xvec=linspace(min(samples),max(samples),1000);
y24=exp(-(xvec.^2)/2 -(xvec.^4)/2);
area=trapz(xvec,y24);
y24norm=y24/area;

```

%From graph we can see that the Metropolis algorithm samples the Boltzmann
 %distribution of the anharmonic oscialltor without any approximation. For
 %sufficiently large t and a suitable delta. In testing of code it seems we
 %need t to be at least 6 orders of magnitude greater than delta to get a
 %plot from which we can give the above conclusion.

```

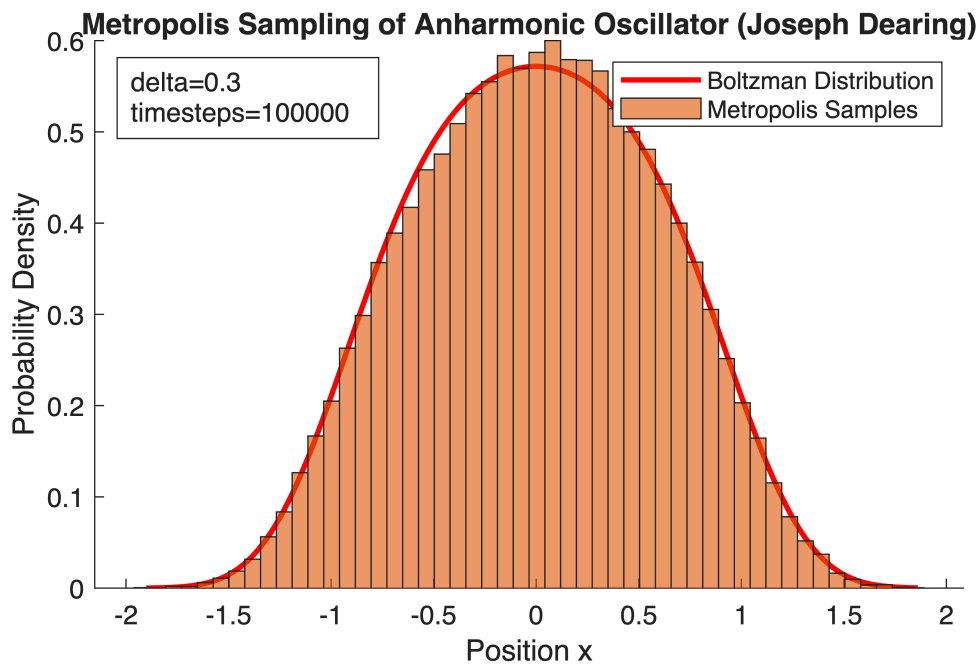
clf;
hold on
plot(xvec, y24norm, 'r-', 'LineWidth', 2)
histogram(samples,50,'Normalization','pdf')
xlabel('Position x')
ylabel('Probability Density')
p={['delta=',num2str(d)],['timesteps=',num2str(t)]};
annotation('textbox',[0.15 0.8 0.1 0.1],'string',p)
title('Metropolis Sampling of Anharmonic Oscillator (Joseph Dearing)')
legend('Boltzman Distribution','Metropolis Samples')
hold off
end

```

```

metropolis(0.3,100000)

```



```
function factormetro(d,t)
x=0;
samples=zeros(1,t);
pi2=@(x)exp(-(x^2)/2);
pi4=@(x)exp(-(x^4)/4);
for i = 1:t
    dx=d*(2*rand()-1);
    r2=pi2(x+dx)/pi2(x);
    r4=pi4(x+dx)/pi4(x);
    upsilon=rand();
    if upsilon<(min(1,r2)*min(1,r4))
        x=x+dx;
    end
    samples(i)=x;
end

xvec=linspace(min(samples),max(samples),1000);
y24=exp(-(xvec.^2)/2 -(xvec.^4)/4);
area=trapz(xvec,y24);
y24norm=y24/area;
```

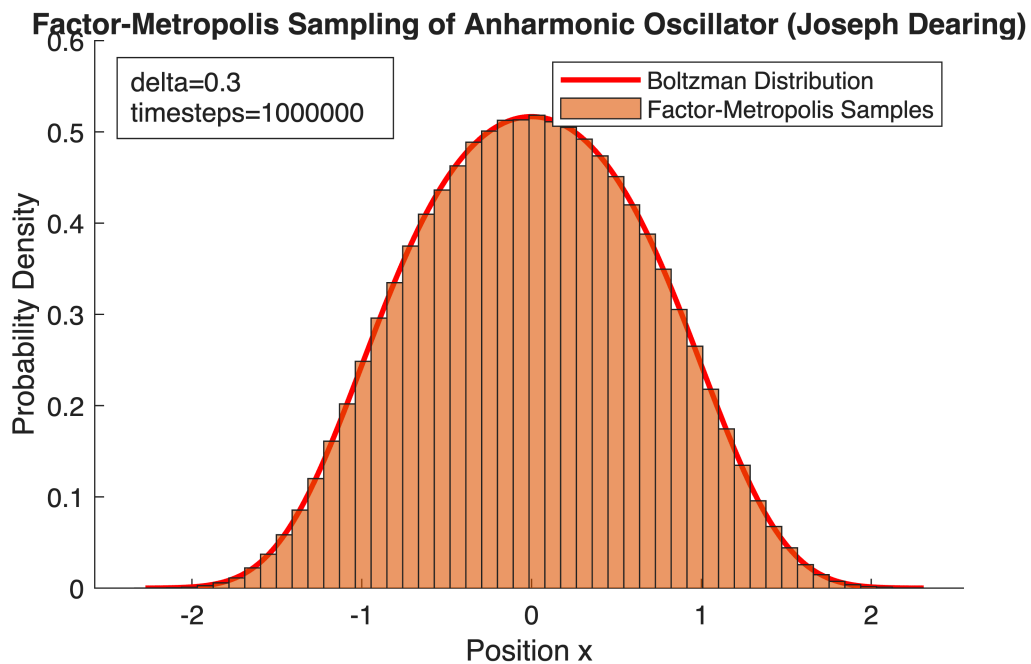
%As before we can see that the algorithm samples the Boltzmann distribution
%of the anharmonic oscillator without any approximation. Furthermore, it
%from testing it seems we need the same difference in order of magnitude
%of
%delta and t to make such conclusion.

```
%pi24(x)=pi2(x)*pi4(x)
%Meaning the Factor-Metropolis algorithm preserves the same detailed  
%balance condition as that for the Metropolis algorithm, hence it  
converges  
%to the Boltzmann distribution of the anharmonic oscillator without any
```

```

%approximation.
clf;
hold on
plot(xvec, y24norm, 'r-', 'LineWidth', 2)
histogram(samples,50,'Normalization','pdf')
xlabel('Position x')
ylabel('Probability Density')
p={['delta=',num2str(d)],['timesteps=',num2str(t)]];
annotation('textbox',[0.15 0.8 0.1 0.1],'string',p)
title('Factor-Metropolis Sampling of Anharmonic Oscillator (Joseph Dearing)')
legend('Boltzman Distribution','Factor-Metropolis Samples')
hold off
end
factormetro(0.3,1000000)

```



```

function liftedmetropolis(d,t)

x=0;
samples=zeros(1,t);

sigma=1;
pi24=@(x)exp(-x^2/2-x^4/4);

for i = 1:t
    dx=d*(2*rand()-1);
    epsilon=rand();

    if epsilon < min(1,pi24(x+sigma*dx)/pi24(x))
        x=x+sigma*dx;
    else
        sigma=-sigma;
    end
end

```

```

samples(i)=x;

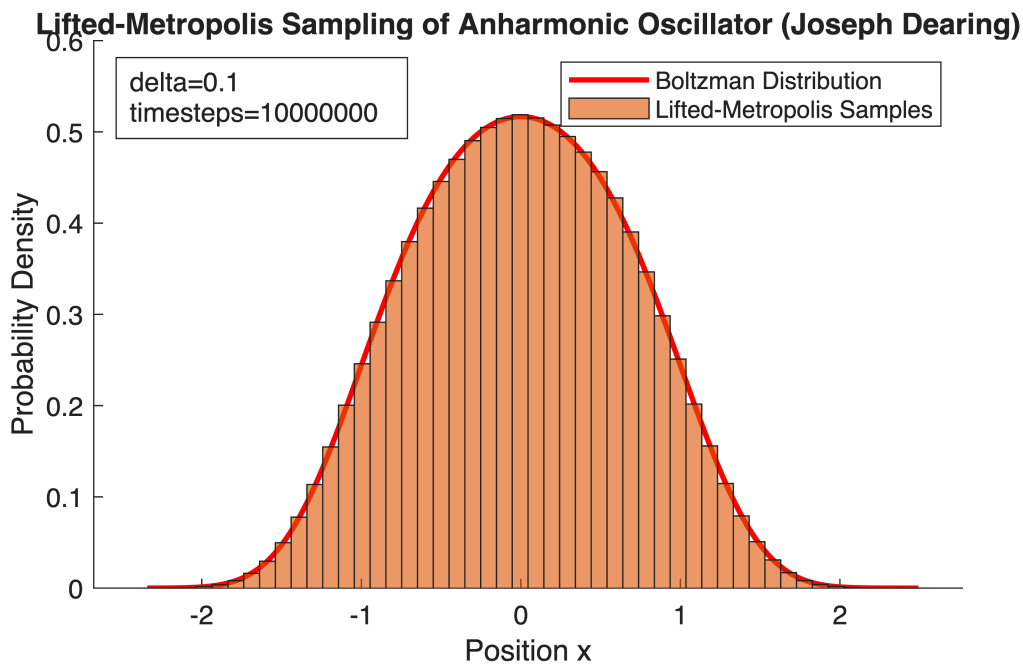
end

xvec=linspace(min(samples),max(samples),1000);
y24=exp(-(xvec.^2)/2 -(xvec.^4)/4);
area=trapz(xvec,y24);
y24norm=y24/area;

clf;
hold on
plot(xvec, y24norm, 'r-', 'LineWidth', 2)
histogram(samples,50,'Normalization','pdf')
xlabel('Position x')
ylabel('Probability Density')
p={['delta=',num2str(d)],['timesteps=',num2str(t)]];
annotation('textbox',[0.15 0.8 0.1 0.1],'string',p)
title('Lifted-Metropolis Sampling of Anharmonic Oscillator (Joseph Dearing)')
legend('Boltzman Distribution','Lifted-Metropolis Samples')
hold off
end

liftedmetropolis(0.1,10000000)

```



```

%Joseph Dearing

function zigzag(n,b) %(number of events,beta)

x=0;
t=0;
sigma=1;
samples=[];

```

```

for i = 1:n
    xev=sigma*sqrt(-1+sqrt(1-4*(b^-1)*log(rand())));
    tev=t+abs(xev-x);

    tstar=floor(t)+1:floor(tev);

    if ~isempty(tstar)
        newsamples=x+sigma*(tstar-t);
        samples=[samples,newsamples];
    end

    x=xev;
    sigma=-sigma;
    t=tev;

end

xvec=linspace(min(samples),max(samples),1000);
y24=exp(-(xvec.^2)/2 -(xvec.^4)/4);
area=trapz(xvec,y24);
y24norm=y24/area;

%From the graph we can clearly see that the metropolis, lifted-metropolis
%and the event driven zig-zag algorithms sample the same distribution
%correctly

%detailed balance equation  $\pi(x)P(x,x')=\pi(x')P(x',x)$  for all  $x,x'$ 
%global balance  $\pi(x)=\sum \text{over } x' [\pi(x')P(x',x)]$ 
%Metropolis satisfies detailed since its a reversible markov chain
%Lifted and Zig-Zag satisfy global since they are non-reversible

clf;
hold on
plot(xvec, y24norm, 'r-', 'LineWidth', 2)
histogram(samples,50,'Normalization','pdf')
xlabel('Position x')
ylabel('Probability Density')
p={['number of events=',num2str(n)],['Beta=',num2str(b)]];
annotation('textbox',[0.15 0.8 0.1 0.1],'string',p)
title('Zig-Zag Sampling of Anharmonic Oscillator (Joseph Dearing)')
legend('Boltzman Distribution','Zig-Zag Samples')
hold off
end

zigzag(100000,1)

```